Chaining

key → 4, 6, 14, 9, 12, 34, 44, 19

$h(k) = k \% m$

$h(4) = 4 \% 10$
$= 4$

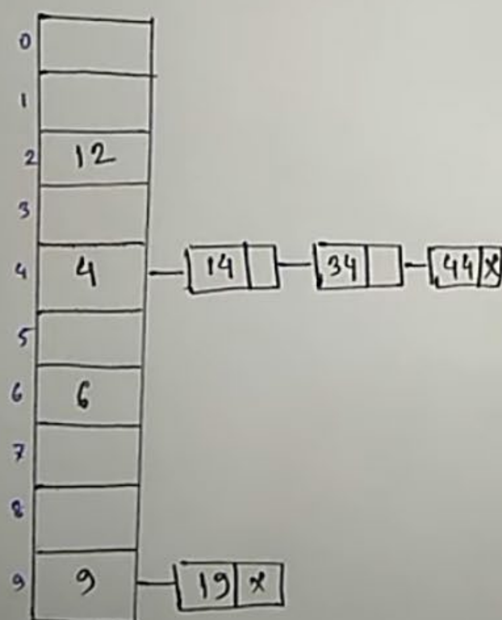
$h(6) = 6 \% 10$
$= 6$

$h(14) = 14 \% 10$
$= 4$

$h(9) = 9 \% 10$
$= 9$

$h(12) = 12 \% 10$
$= 2$

$h(34) = 34 \% 10$
$= 4$



Linear Probing

key → 4, 6, 14, 9, 12, 34

$h'(k) = [h(k) + i] \%$

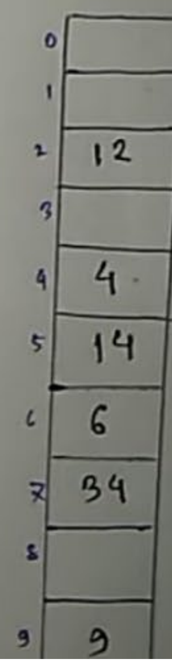
$h(k) = k \% m$

$h(14) = 14 \% 10$
$= 4$

$h'(14) = [4 + 0] \% 10$
$= 4$
$= [4 + 1] \% 10$
$= 5 \% 10$
$= 5$

$h(34) = 34 \% 10$
$= 4$

$h'(34) = [4 + 0] \% 10$



key → 4, 6, 14, 9, 12, 34, 44, 19

$h(k) = k \% m$

$h(4) = 4 \% 10$
$= 4$

$h(6) = 6 \% 10$
$= 6$

$h(14) = 14 \% 10$
$= 4$

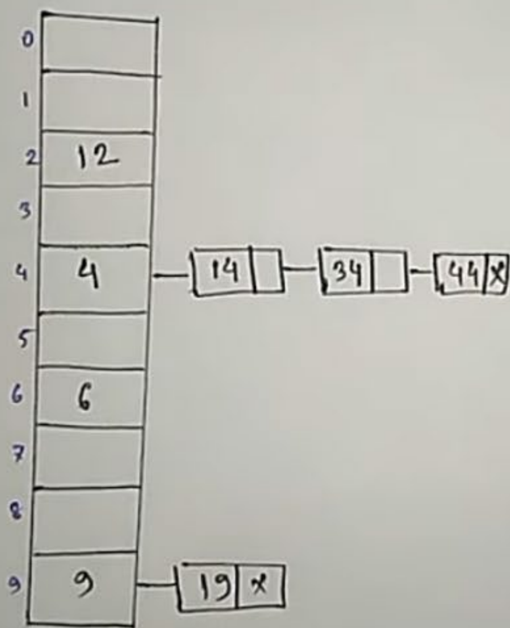
$h(9) = 9 \% 10$
$= 9$

$h(12) = 12 \% 10$
$= 2$

$h(34) = 34 \% 10$
$= 4$

$h(44) = 44 \% 10$
$= 4$

$h(19) = 19 \% 10$

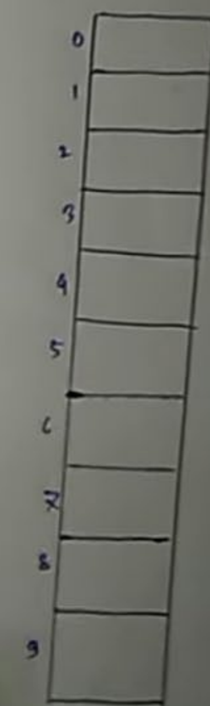


Linear Probing

key → 4, 6, 14, 9, 12

$h'(k) = h(k$

$h(k) = k \% m$



% m

% 10

4

% 10

4 % 10

4

9 % 10

= 9

= 12 % 10

= 2

= 34 % 10

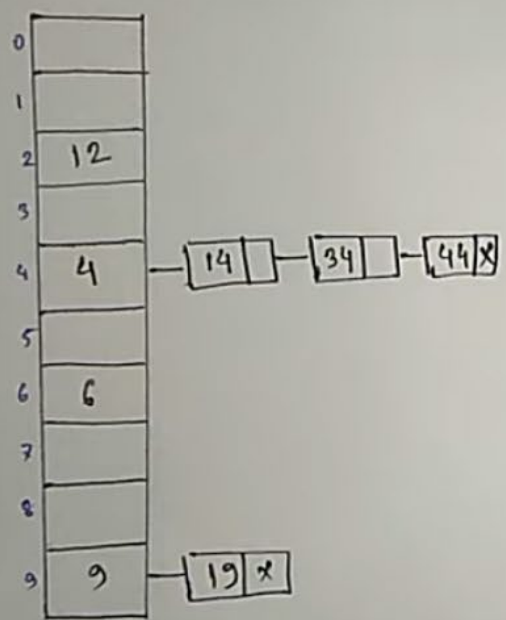
= 4

= 44 % 10

= 4

) = 19 % 10

= 9



key → 4, 6, 14, 9, 12, ...

$h(k) = k \% m$

$h(14) = 14 \% 10$

$= 4$

$h'(14) = [4 + 0] \% 10$

$= 4$

$= [4 + 1] \% 10$

$= 5 \% 10$

$= 5$

$h(34) = 34 \% 10$

$= 4$

$h'(34) = [4 + 0] \% 10$

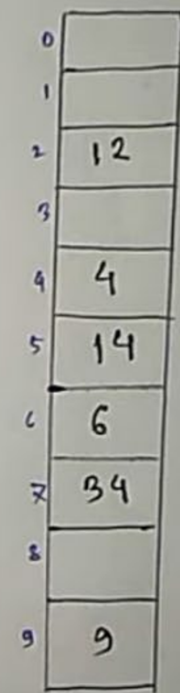
$= 4$

$= [4 + 1] \% 10 = 5$

$= [4 + 2] \% 10 = 6$

$= [4 + 3] \% 10 = 7$

$=$



Collision Resolve

Quadratic Probing

$h'(k) = [h(k) + i^2] \% m$

key → 4, 6, 14, 9, 12, 34

$h(k) = k \% m$

$h(14) = 14 \% 10$

$= 4$

$h'(14) = [4 + 0] \% 10 = 4$

$= [4 + 1] \% 10 = 5$

0	
1	
2	
3	
4	4
5	14
6	6
7	
8	
9	

$h'(k) = [h(k) + i^2] \% m$

key → 4, 6, 14, 9, 12, 34

$h(k) = k \% m$

$h(14) = 14 \% 10$

$= 4$

$h'(14) = [4 + 0] \% 10 = 4$

$= [4 + 1] \% 10 = 5$

$h(34) = 34 \% 10 = 4$

$h'(34) = [4 + 0] \% 10 = 4$

$= [4 + 1] \% 10 = 5$

$= [4 + 4] \% 10 = 8$

0	
1	
2	12
3	
4	4
5	14
6	6
7	
8	34
9	9

Double Hashing

$h_1(k) = k \% m$

$h_2(k) = k \% 8$

$h_3(k) = [h_1(k) + i * h_2(k)] \% m$

key → 4, 6, 14, 9, 12, 34, 44, 72, 54

$h_1(4) = 4 \% 10 = 4$

$h_1(6) = 6 \% 10 = 6$

$h_1(14) = 14 \% 10 = 4$

$h_2(14) = 14 \% 8 = 6$

$h_3(14) = [4 + 0 * 6] \% 10$

$= 4 \% 10 = 4$

$= [4 + 1 * 6] \% 10$

$= 10 \% 10 = 0$

0	14
1	
2	
3	
4	4
5	
6	6
7	
8	
9	

$h_1(6) = 6 \% 10 = 6$

$h_1(14) = 14 \% 10 = 4$

$h_2(14) = 14 \% 8 = 6$

$h_3(14) = [4 + 0 * 6] \% 10$

$= 4 \% 10 = 4$

$= [4 + 1 * 6] \% 10$

$= 10 \% 10 = 0$

$h_1(9) = 9 \% 10 = 9$

$h_1(12) = 12 \% 10 = 2$

$h_1(34) = 34 \% 10 = 4$

$h_2(34) = 34 \% 8 = 2$

$h_3(34) = [4 + 0 * 2] \% 10$

$= 4 \% 10 = 4$

$= [4 + 1 * 2] \% 10$

$= 6 \% 10 = 6$

$= [4 + 2 * 2] \% 10$

$= 8 \% 10 = 8$

0	14
1	
2	12
3	
4	4
5	
6	6
7	
8	34
9	9

Cases when elements will not be inserted in the hash table

$h_3(k) = [\,h_1(k) \ldots$

$h_1(4) = 4 \% 10 = 4$

$h_1(6) = 6 \% 10 = 6$

$h_1(14) = 14 \% 10 = 4$

$h_2(14) = 14 \% 8 = 6$

$h_3(14) = [4 + 0 * 6] \% 10$

$= 4 \% 10 = 4$

$= [4 + 1 * 6] \% 10$

$= 10 \% 10 = 0$

$h_1(9) = 9 \% 10 = 9$

$h_1(12) = 12 \% 10 = 2$

$h_1(34) = 34 \% 10 = 4$

$h_2(34) = 34 \% 8 = 2$

$h_3(34) = [4 + 0 * 2] \% 10$

$= 4 \% 10 = 4$

$= [4 + 1 * 2] \% 10$ | $= [4 + 2 * \ldots$

0	14
1	
2	12
3	
4	4
5	
6	6
7	
8	34
9	[illegible]

$h_1(44) = 44 \% 10 = 4$

$h_2(44) = 44 \% 8 = 4$

$h_3(44) = [4 + 1 * 4] \% 10$

$= 8 \% 10 = 8$

$= [4 + 2 * 4] \% 10$

$= 12 \% 10 = 2$

$= [4 + 3 * 4] \% 10$

$= 16 \% 10 = 6$

$= [4 + 4 * 4] \% 10$

$= 20 \% 10 = 0$

$= [4 + 5 * 4] \% 10$

$= 24 \% 10 = 4$

$= [4 + 6 * 4] \% 10$

$= 28 \% 10 = 8$

$= 4 \% 10 = 4$

$= [4 + 1 * 2] \% 10$

$= 6 \% 10 = 6$

$= [4 + 2 * 2] \% 10$

$= 8 \% 10 = 8$

$= [4 + 7 * 4] \% 10$

$= 32 \% 10 = 2$

$= [4 + 8 * 4] \% 10$

$= 36 \% 10 = 6$

$[4 + 9 * 4] \% 10$

$= 40 \% 10 = 0$

farhanhasan[illegible]@gmail.com

$h_1(72) = 72 \% 10 = 2$

$h_2(72) = 72 \% 8 = 0$

$h_3(72) = [2 + 0 * 0] \% 10$

$= 2 \% 10 = 2$

$= [2 + 1 * 0] \% 10$

$= 2 \% 10 = 2$

54

$h_1(54) = 54 \% 10 = 4$

$h_2(54) = 54 \% 8 = 6$

$h_3(54) = [4 + 1 * 6] \% 10$

$= 10 \% 10 = 0$

10.3.3 Random probing method

Here the following function may be used as probing function:

$y_{i+1} = (y_i + r) \bmod m$, where $i = 0, 1, 2, 3$ and so on and r is an integer value that is relatively prime to m. Two integers are relatively prime to each other if their greatest common divisor is 1, *i.e.*,

gcd $(r, m) = 1$; //gcd for Greater Common Divisor.

If $y_0 = 3$, and $r = 2$, then $y_1 = 5$, $y_2 = 7$, $y_3 = 9$, $y_4 = 11$ etc.